# A Self-Supervised Dual-Modality Work-frame for Breast Cancer Detection via Thermography and Mammography with Gated Fusion

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**Abstract.** Infrared thermography offers a radiation-free, low-cost alternative to mammographic X-ray screening for breast cancer detection; however, its restriction to surface-level thermal signals and the scarcity of well-structured publicly available labeled images limit its reliability as a stand-alone diagnostic tool—motivating this study to designate thermography as the primary modality while treating mammography as a secondary cross-domain reference. A VICReg-based self-supervised work-frame is proposed, employing a modified ResNet-50 with Gabor texture enhancement, bilateral symmetry regularization, and a Gated Fusion mechanism combining dual-pooled spatial descriptors with SSL embeddings. The thermography dataset is split at the *patient level*—a rigour absent from most prior thermographic studies—and under these conditions the work-frame attains 71.60% accuracy on malignant vs. benign classification (malignant recall: 0.86, F1: 0.7526, AUC: 0.7816), surpassing all non-SSL baselines on both metrics. On the INbreast mammography dataset (balanced), the identical architecture achieves 98.24% accuracy and AUC of 0.9987, confirming cross-modal generalization.

**Keywords:** Breast cancer detection · Thermography · Self-supervised learning · Gated fusion · Mammography · VICReg · SSL · dual-modality

## 1 Introduction

**Background.** Breast cancer remains a leading cause of cancer mortality in women worldwide [11][12]. Mammography, the established screening standard for breast-cancer detection, depends on ionizing X-ray radiation that poses cumulative exposure risks [13] and comes with high costs that restrict access in resource-limited areas. Infrared thermography offers a radiation-free, low-cost alternative to it that captures thermal emissions linked to tumor-driven angiogenesis [14]. Despite renewed interest due to this modality's compatibility with deep learning (DL) classifiers [10], thermography remains unapproved as a standalone diagnostic tool as it is constrained to capture only surface-level signals, while, in image processing, it is hampered by the scarcity of annotated and well structured datasets [5][10]. SSL has been shown to improve classification across mammographic and ultrasound imaging [5], and a small number of studies have begun applying it to thermographic images [3][8], though fundamental gaps remain.

**Problem Statement.** Existing SSL-based breast cancer detection studies predominantly target mammography or ultrasound [1][2][5][7], with very few addressing thermography [3][8]. Roslidar et al.[3] achieved 84.37% top-1 accuracy via SimCLR on DMR-IR but lacked patient-level splitting—risking data leakage—and their ∼2 GB model is impractical for constrained hardware. Saha et al. [8] applied SSL to 3D thermal reconstruction without addressing malignancy classification. Supervised-only thermography studies report either inflated accuracies under non-rigorous splits [9] or limited benign/malignant discrimination [6]. Cross-modal generalization of a single SSL architecture also remains unexplored.

**Contributions.** This work proposes a VICReg-based dual-modality SSL framework designating thermography as the primary modality while validating cross-domain generalizability on mammography. Key contributions: **(i)** a modified ResNet-50 with Gabor texture enhancement and bilateral symmetry regularization for thermal imaging; **(ii)** a Gated Fusion module combining dual-pooled spatial descriptors with SSL embeddings; **(iii)** strict patient-level splitting for thermography [3, 9] eliminating data leakage; and **(iv)** performance surpassing all non-SSL baselines—71.60% accuracy (recall: 0.86, F1: 0.7526, AUC: 0.7816) on thermography and 98.24% accuracy (AUC: 0.9987) on INbreast mammography.

## 2 Related Works

Wang et al.'s [5] Review Article on Self-supervised learning and transformer-based technologies in breast cancer imaging (reviewed 85 articles) mostly focusing on breast cancer, breast tumour, and breast lesions related studies that used SSL in their studies. The reviewed studies mostly used Ultrasound images and Mammographic images; some used histopathological images. None of their reviewed article used SSL on thermographic images which raises question on the practise of thermographic images. However, their study collectively reports that SSL improves classification accuracy though the magnitude of improvement varies. They also reported the hurdles regarding the quadratic complexity relative to input size which limits real-time inference in clinical settings without access to high-performance GPUs.

Mashekova et al.'s [10] Review of Artificial Intelligence Techniques for Breast Cancer Detection with Different Modalities: Mammography, Ultrasound, and Thermography Images, examines the availability and use of different modality of images for breast cancer detection purpose. Their study indicates that though thermographic images are used for studies in breast cancer detection field, it is still not approved as a medical tool; it can be used as an adjunct tool though as it comes with the limitation of capturing only surface-level measurements, thereby excluding the diagnosis of tumours located deeper within the tissues. However, studies continue using this modality of images as it reduces reliance on X-rays (mostly used during mammography), and thermographic image processing also comes with lower cost. Also, despite being historically underutilized, this modality has gained renewed interest due to its compatibility with DL classifiers. Now their review includes 9 articles (experimented ML, DL, Hybrid Algorithms) that used thermographic images in their studies. Five of them used CNN architectures; 1 used VGG-19. Most of them used the known DMR-IR (Database for Mastology Research with Infrared Image) dataset.

Most of the studies in this field use Supervised classification for cancer detection. Attallah et al.'s [6] Thermo-CAD used CNN with NNMF and Relief-F on thermographic images, achieving 100% accuracy (normal vs. abnormal, indicating overfitting) and 79.3% (CSVM) on benign/malignant—the same dataset as ours—with poor malignant recall (0.5429) and no class balancing. Alshehri et al. [9] reported attention-based CNN test accuracies of 99.46%–99.30% on DMR-IR, though without external validation and with overfitting risk.

One of the earliest studies using SSL specifically on thermographic images was 3D-BreastNet [8] by Saha et al. in 2023, which reconstructed 3D breast surfaces from 2D thermal images using SSL (Otsu segmentation), achieving accuracy/Dice/Jaccard/Hausdorff of 0.97/0.97/0.94/3.86 on 1,000 images from 200 patients with maintained patient-level separation; however, no malignancy classification was integrated. After that, Roslidar et al.'s [3] bi-pipeline SSL model applied SimCLR with ResNet-50 on DMR thermographic images, achieving 84.37% top-1 accuracy on 650 images, but without patient-level splitting (risking data leakage) and with high computational cost ($\approx$2 GB total model size).

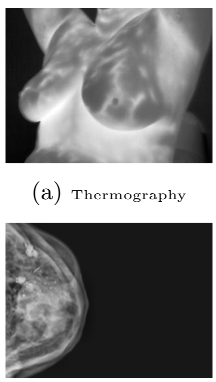
On the mammography field, Chen et al.'s [1] HybMNet (Swin-T + CNN, SSL) on CMMD and INbreast showed SSL improving F1 (0.678$\rightarrow$0.719) and accuracy (0.753$\rightarrow$0.830), though not exceeding 80%. Cai et al.'s [2] ViT-GNN on CBIS-DDSM achieved accuracy 84.2%, F1 0.84, but without patient-level splits and with potential ImageNet pre-training bias. Mohammadi et al.'s [7] TSSL-MSViT achieved AUCs of 0.967 (CBIS-DDSM) and 0.972 (INbreast) via Multi-Scale Masked Reconstruction and Cross-Scale Contrastive Learning, consistently outperforming CNN and ViT baselines; however, the focus on AUC rather than class-level F1 leaves benign/malignant discrimination under-examined.

Huang et al.[4], on the other hand, chose a different kind of framework called Flip Learning, a Weakly Supervised Segmentation technique using reinforcement-based learning on breast ultrasound images, achieving DICE$\uparrow$ and JAC$\uparrow$ scores of $92.37 \pm 4.22$, $86.09 \pm 6.80$ on two test sets; however, their framework is nodule-segmentation based and not specifically designed for cancer classification yet.

Our study primarily targets thermography—constrained by its surface-level limitations and data scarcity—while treating mammography as a cross-domain validation reference.

## 3 Methodology

Our study proposes a dual-modality automated breast cancer detection framework for breast thermography and digital mammography. As each modality has its own unique characteristics (thermography captures surface-level infrared intensity patterns where cancerous regions produce irregular high-frequency thermal patterns; mammography captures whole structural characteristics with black background and white foreground),



(a) Thermography

(b) Mammography

Fig. 1: Breast Thermography & Mammography

each modality has its own preprocessing and learning pipeline tailored to the imaging medium's physical properties, before converging into a shared SSL and supervised architecture. The design addresses scarce labeled medical data, class imbalance, and inter-modality signal differences while maintaining a patient-level split for the thermography setting.

### 3.1 Dataset and Preprocessing

The thermographic framework was trained on a primary dataset [15] of 357 images from 119 patients maintaining patient-level split, and the INbreast dataset [16] of 7,632 mammogram images was used for the mammographic setting of the framework.

**Dataset Organization, Splitting, and Modality-Specific Preprocessing:**

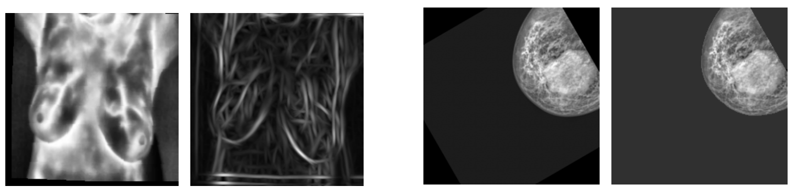
- **Thermography:** Images were split at the *patient* level (as in same patient image is not found across train, val, or test fraction of the split, as our thermography dataset has 3 angled images per same patient), preventing thermal patterns from the same patient leaking across fractions (i.e., train, val, test)—a rigour absent from most of the prior thermographic studies. Stratified partitioning yielded ≈285 training, 36 validation, and 36 test patients (Benign/Malignant). The labeled training split served dually as the unlabeled SSL pretraining pool and the supervised fine-tuning set; the raw Unlabeled category was excluded.
  Preprocessing resizes images to $224 \times 224$ grayscale, applies a $3 \times 3$ median filter to preserve fine thermal gradients, then CLAHE (clip limit 2.0, $8 \times 8$ grid) to amplify localized hyperthermia. Brightness/contrast augmentations are excluded to protect the diagnostic temperature signal.
  **Gabor Filter-Based Texture Enhancement (Thermography Only):** A 24-kernel Gabor filter bank (8 orientations $\theta \in [0, \pi)$; $\sigma \in \{2, 4, 6\}$; size $21 \times 21$; $\lambda$=10, $\gamma$=0.5) enhances oriented vascular texture features prior to SSL. The 2D Gabor kernel is:

$$g(x, y;\, \lambda, \theta, \sigma, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cos\left(\frac{2\pi x'}{\lambda}\right) \tag{1}$$

*Eq. 1: oriented Gabor kernel capturing frequency- and scale-selective thermal texture.* where $x' = x\cos\theta + y\sin\theta$ and $y' = -x\sin\theta + y\cos\theta$. The maximum response across all kernels is retained and normalized to $[0, 255]$:

$$I_{\text{Gabor}}(x, y) = \text{Normalize}\left(\max_k \left|I * g_k\right|(x, y)\right), \quad k = 1, \ldots, 24 \tag{2}$$

*Eq. 2: max-pool response across all 24 kernels, yielding a single-channel enhanced texture map.*



(a) CLAHE vs Gabor bank: Thermography (b) Original vs preprocessed: Mammography

Fig. 2: Applied preprocessing for both modalities

- **Mammography:** The INbreast dataset was split at image level via stratified sampling (80/10/10), yielding 6,096 training, 762 validation, and 762 test images before augmentation.
  Images are resized to $224 \times 224$ grayscale. Per-image z-score normalization reduces scanner-specific bias:

$$\hat{I}(x, y) = \frac{I(x, y) - \mu}{\sigma + \varepsilon} \tag{3}$$

*Eq. 3: per-image z-score normalization removing scanner-specific intensity bias.* where $\mu = \text{mean}(I)$, $\sigma = \text{std}(I)$, and $\varepsilon = 1 \times 10^{-8}$.
Otsu's thresholding isolates breast tissue by maximizing between-class variance:

$$\sigma_B^2(t) = \omega_0(t)\,\omega_1(t)\big[\mu_0(t) - \mu_1(t)\big]^2, \quad T^* = \arg\max_t \sigma_B^2(t) \tag{4}$$

*Eq. 4: Otsu's optimal threshold maximising between-class variance for background/tissue separation.* Morphological closing then opening ($5 \times 5$ structuring element) fill gaps and remove spurious regions. The largest connected component is retained as binary mask $M$, and the segmented image is:

$$I_{\text{seg}}(x, y) = \hat{I}(x, y) \times M(x, y), \qquad M(x, y) \in \{0, 1\} \tag{5}$$

*Eq. 5: binary mask retaining only diagnostically relevant breast tissue.* Pixel values are then rescaled to $[0, 255]$ and saved as lossless PNGs.

**Class Balancing and Dataset Expansion: Mammography:** ROS with Random Affine Transformations (rotation ±15°, translation/scale ±10%) corrected a 2,064-image Malignant deficit. Final balanced dataset: 8,160 training (4,080 per class), 1,020 validation, 1,020 test.

**Thermography:** Each image was expanded to a triplet (original, horizontal flip, stochastic affine). Vertical flips were excluded as anatomically implausible. Final counts: 1,206 training (603 per class, ROS-balanced), 144 validation, 162 test.

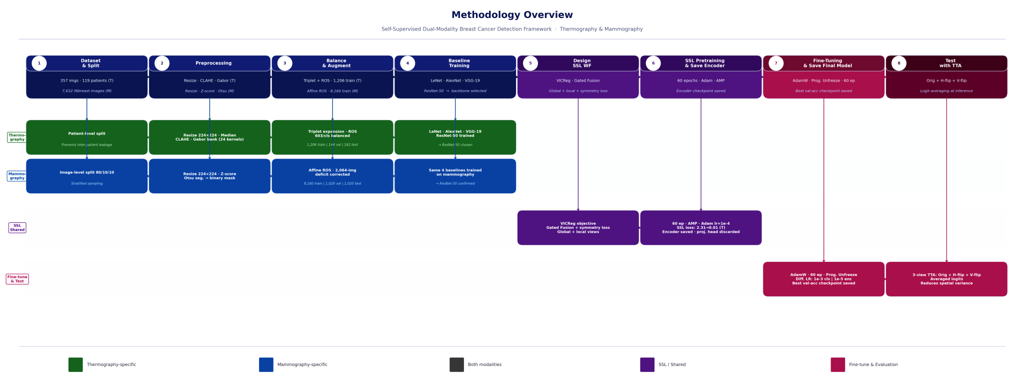



Fig. 3: Methodology Diagram: all steps followed to develop the proposed work-frame/model

### 3.2 Self-Supervised Pretraining

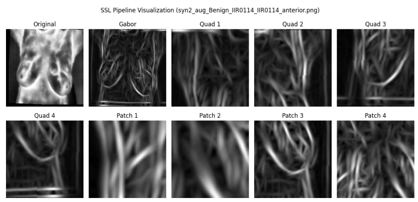
SSL pretraining preceded supervised fine-tuning for both modalities, using a modified ResNet-50 accepting single-channel grayscale input (`Conv2d(1, 64, kernel=`$7 \times 7$`, stride=2, padding=3)`). The VICReg objective promotes invariance across augmented views, variance across the batch, and non-redundant embedding dimensions. Given projected embeddings $Z$ and $Z'$ of shape $(N, D)$:

$$\mathcal{L}_{\text{sim}} = \frac{1}{ND} \sum_i \sum_j (Z_{ij} - Z'_{ij})^2 \tag{6}$$

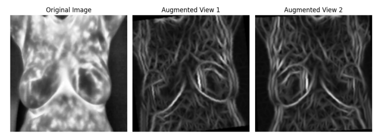
*Eq. 6: invariance loss penalising divergence between embeddings of two augmented views.*

$$\mathcal{L}_{\text{var}} = \frac{1}{D} \sum_j \max\bigl(0,\, 1 - \sqrt{\text{Var}(Z_j)}\bigr) + \frac{1}{D} \sum_j \max\bigl(0,\, 1 - \sqrt{\text{Var}(Z'_j)}\bigr) \tag{7}$$

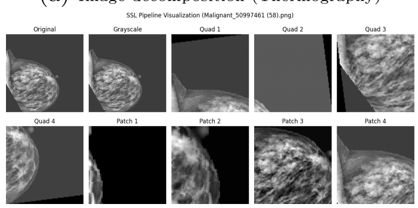
*Eq. 7: variance loss preventing representation collapse by enforcing per-dimension spread* $\geq 1$. The covariance term was omitted ($\mathcal{L}_{\text{VICReg}} = \mathcal{L}_{\text{sim}} + \mathcal{L}_{\text{var}}$). The total pretraining objective integrates global, local, and (for thermography) symmetry



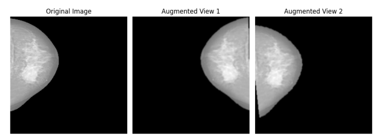

(a) Image decomposition (Thermography)




(b) Augmented pair (global view, thermography)




(c) Image decomposition (Mammography)




(d) Augmented pair (global view, mammography)

Fig. 4: Image decomposition in SSL workflow for global, quadratic, and four random patch views [left]; augmented pairs for contrastive learning (global view) [right].

terms:

$$\mathcal{L}_{\text{SSL}} = \mathcal{L}_{\text{global}} + \frac{1}{K}\sum_{k}\mathcal{L}_{\text{local},k}\ [+\mathcal{L}_{\text{sym}}]_{\text{thermography only}} \tag{8}$$

*Eq. 8: total SSL objective combining global, local-region, and (thermography only) symmetry losses.* where $\mathcal{L}_{\text{global}}$ is VICReg over two full-image views, $\mathcal{L}_{\text{local},k}$ is VICReg over $K$=8 sub-regions (4 quadrant pairs + 4 patch pairs at 56×56 and 112×112), and $\mathcal{L}_{\text{sym}}$ is the thermography-only bilateral symmetry regularizer. The symmetry loss minimizes MSE between each encoder feature map $F \in \mathbb{R}^{B\times C\times H\times W}$ and its horizontal mirror:

$$\mathcal{L}_{\text{sym}} = \frac{1}{BCHW}\sum_{b,c,h,w}\big[F(b,c,h,w) - F(b,c,h,W{-}1{-}w)\big]^2 \tag{9}$$

*Eq. 9: bilateral symmetry regulariser penalising asymmetry in thermographic feature maps.* Pretraining ran for 60 epochs (Adam, $\text{lr} = 10^{-4}$; effective batch size 32 via 4-step gradient accumulation; AMP). The SSL loss converged from 0.443 to ≈ 0.00024 for mammography and from ≈ 2.31 to ≈ 0.01 for thermography.

### 3.3 Network Architecture and Gated Fusion

After pretraining, adaptive average and max pooling are applied independently to the 7 × 7 feature maps and concatenated into a 4,096-dimensional spatial descriptor. A Gated Fusion module then dynamically combines this descriptor with the 512-dimensional SSL projection:

$$g = \sigma\Big(W_2\,\text{ReLU}\big(W_1[f_{\text{sp}}\,;\,f_{\text{ssl}}] + b_1\big) + b_2\Big) \tag{10}$$

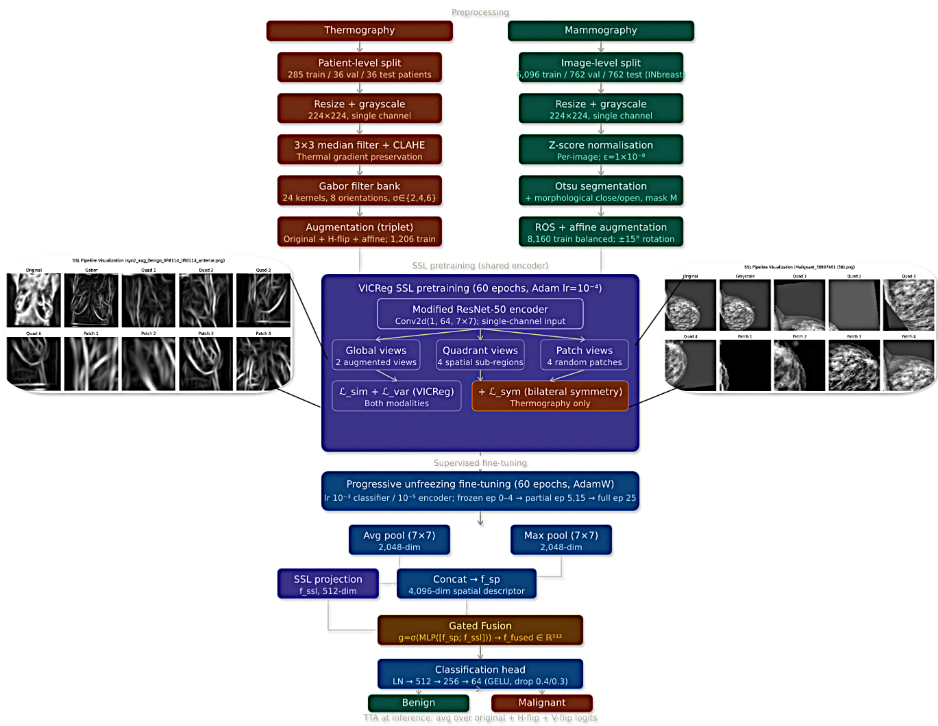



Fig. 5: SSL dual modality architecture (generic) ***[Proposed model]***

*Eq. 10: gating vector $g$ learned from the concatenation of spatial and SSL features.*

$$f_{\text{fused}} = g \odot W_{\text{sp}}\, f_{\text{sp}} + (1 - g) \odot W_{\text{ssl}}\, f_{\text{ssl}} \tag{11}$$

*Eq. 11: gated interpolation between spatial descriptor and SSL embedding into a 512-d fused representation.* where $f_{\text{sp}} \in \mathbb{R}^{4096}$, $f_{\text{ssl}} \in \mathbb{R}^{512}$, $\sigma$ is sigmoid, $\odot$ is element-wise multiplication, and $W_1 \in \mathbb{R}^{512\times 4608}$, $W_2 \in \mathbb{R}^{512\times 512}$ are learned gating MLP projection matrices. The output $f_{\text{fused}} \in \mathbb{R}^{512}$ is passed to a classification head: LayerNorm $\rightarrow$ 512$\rightarrow$256$\rightarrow$64 with GELU and decreasing dropout (0.4, 0.3) $\rightarrow$ binary output.

### 3.4 Supervised Fine-Tuning Setup

Fine-tuning used AdamW for 60 epochs with differential learning rates ($10^{-3}$ for the classifier and fusion module; $10^{-5}$ for the encoder) and a progressive unfreezing schedule: encoder frozen for epochs 0–4; final residual block unfrozen at epoch 5; penultimate block at epoch 15; full encoder at epoch 25. The classifier learning rate followed warmup then cosine annealing:

$$\lambda_{\text{cls}}(t) = \begin{cases} \dfrac{t+1}{W}, & \text{if } t < W \\ \dfrac{1}{2}\left(1 + \cos\dfrac{\pi(t-W)}{T-W}\right), & \text{otherwise} \end{cases} \tag{12}$$

*Eq. 12: classifier LR schedule—linear warmup for $W$ epochs then cosine annealing.* where $T$=60 and $W$=5. The encoder followed a slower cosine schedule without warmup:

$$\lambda_{\text{enc}}(t) = \frac{1}{2}\left(1 + \cos\frac{\pi t}{2T}\right) \tag{13}$$

*Eq. 13: encoder LR—slower cosine decay protecting pretrained SSL representations.* The supervised objective was standard binary cross-entropy:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i}\sum_{j\in\{0,1\}} y_{ij}\log p(y=j \mid x_i) \tag{14}$$

*Eq. 14: binary cross-entropy loss for supervised malignant/benign classification.* At inference, Test-Time Augmentation (TTA) averaged logit predictions over three views: original, horizontal flip, and vertical flip, per image.

### 3.5 Key Design Rationale

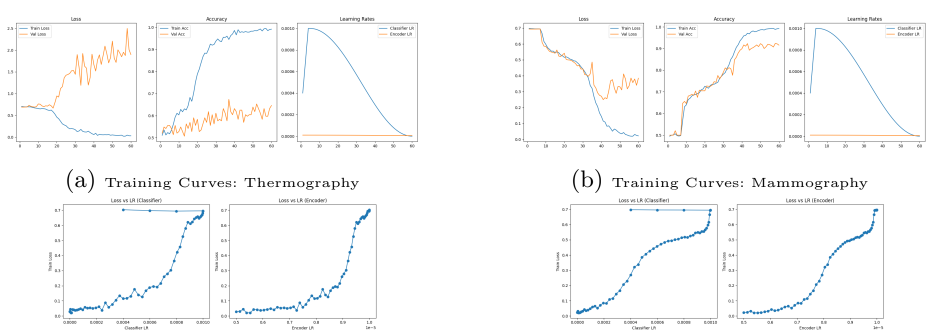
Several deliberate design choices have shaped both pipelines. Single-channel input was maintained throughout the framework, as both datasets are fully grayscale. Patient-level split for thermography dataset was maintained to prevent any temperature-information leakage for a patient. Otsu-based segmentation in the mammography pipeline was chosen to isolate clinically relevant tissue from scanner artifacts, while Gabor texture enhancement in the thermography pipeline captures vascular thermal gradients absent from raw preprocessed images. Use of the same train data for Self-supervised pretraining as well as for the fine tuning part is motivated by the scarcity of well structured Thermography datasets. The progressive unfreezing schedule was chosen to protect the pretrained representations from premature overwriting by fine tuning for the small dataset (this setting was designed keeping the small thermography dataset in consideration). Together, these choices reflect a methodology built around the specific diagnostic and statistical properties of each imaging modality.

## 4 Discussion

### 4.1 Self-Supervised Pretraining and Training Dynamics

The SSL pretraining phase converged consistently over 60 epochs. For mammography, the VICReg loss dropped from 0.443 (epoch 0) to below 0.001 (by epoch 22), and stabilized near 0.00024 by epoch 59 which indicates rapid structural learning followed by gradual fine-grained refinement. The gradient accumulation strategy (effective batch 32 by accumulation of physical batch 8) provided stable gradient estimates throughout. For thermography, the loss decreased gradually from ≈2.31 to ≈0.01; the higher initial value is due to the lower resolution of thermal patterns and the additional symmetry loss term for thermography. The bilateral symmetry regularizer guided the thermography encoder to learn the asymmetry between the left and the right breasts for malignant-patient images without any labeled supervision.

During supervised fine-tuning, the progressive unfreezing schedule produced remarkably different training dynamics compared to end-to-end fine-tuning. Mammography training accuracy rose from ≈49.5% to over 99%, with the sharpest gains between epochs 25–40 when the full encoder was unfrozen (peak validation accuracy 92.35% at epoch 57). The thermography model was more gradual, with validation accuracy peaking at 67.36% at epoch 36 before the possible overfitting consistent with the limited 1,206-image training set. However, the final model saved rather than the best model(saved by validation f1-score) gave better results, suggesting the model tool a little more time to get stable and generalized. The choice for differential learning rates for ($10^{-3}$, classifier vs. $10^{-5}$, encoder) were critical: keeping the encoder rate an order of magnitude lower than that of the classifier protected SSL-learned representations from being overwritten too early; this setting was especially vital for the thermography setting, where the small labeled validation set makes aggressive updating of the encoder early destructive for our goal of classification.



(a) Training Curves: Thermography

(b) Training Curves: Mammography

(c) LR Curves: Thermography

(d) LR Curves: Mammography

Fig. 6: Training & LR curves for SSL work-frame for Thermography & Mammography

### 4.2 Gated Fusion and Test-Time Augmentation

The Gated Fusion module in the framework improved classification for both modality by dynamically balancing the 4,096-dimensional dual-pooled spatial descriptor (localized texture) against the 512-dimensional SSL projection (globally coherent semantics) on a per-image basis. For mammography, the gate emphasized between learned spatial feature differences from the local morphological details (margin irregularity, spiculation, microcalcification clustering) and the holistic pattern differences in SSL embeddings. For thermography, it selected the weights between fine thermal gradients and broader anatomical SSL representations.

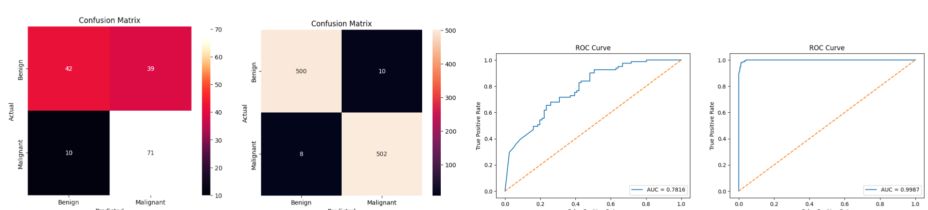
TTA (Test-Time Augmentation; averaging original, horizontal-flip, and vertical-flip predictions of the same image) further improved stability, reducing variance from minor spatial asymmetries; this particularly benefited for thermography, where posture and sensor-distance variations could shift borderline predictions unpredictably.

## 5 Results Analysis

The chosen Baseline models include LeNet-5, AlexNet, VGG-19, and ResNet-50—spanning shallow to deep supervised CNNs respectively, all trained without the SSL structure. Our framework used ResNet-50 (selected based on the baseline model trainings and evaluation metrics [f1-score]) as the backbone which enabled a direct comparison that isolates the contribution of SSL pretraining and Gated Fusion over identical architecture.

### 5.1 Classification Performance Metrics

**Thermography** The thermography (proposed) model achieved test accuracy **71.60%**, F1-score **0.7526**, and AUC **0.7816** across 162 test images. While lower than mammography results, these must be interpreted in the context of the dataset's limited scale and the complexity of thermographic diagnosis.




(a) Confusion Matrix: Thermography (b) Confusion Matrix: Mammography (c) ROC Curve: Thermography (d) ROC Curve: Mammography

Fig. 7: Confusion Matrices & ROC curves for SSL work-frame for Thermography & Mammography

Table 1: Workflow performance comparison with baseline models (thermography).

| Model | Precision | | Recall | | F1-score (pos.) | Acc. (%) |
|---|---|---|---|---|---|---|
| | Malignant | Benign | Malignant | Benign | | |
| LeNet5 | 0.65 | 0.68 | 0.70 | 0.62 | 0.6745 | 66.05 |
| AlexNet | 0.65 | 0.60 | 0.52 | 0.72 | 0.5753 | 61.73 |
| VGG19 | 0.62 | 0.62 | 0.62 | 0.62 | 0.6173 | 61.73 |
| ResNet50 | 0.70 | 0.68 | 0.67 | 0.72 | 0.6835 | 69.13 |
| **Proposed SSL work-frame** | **0.67** | **0.81** | **0.86** | **0.57** | **0.7526** | **71.60** |

Table 2: Workflow performance comparison with baseline models (mammography).

| Model | Precision | | Recall | | F1-score (pos.) | Acc. (%) |
|---|---|---|---|---|---|---|
| | Malignant | Benign | Malignant | Benign | | |
| LeNet5 | 0.98 | 1.00 | 1.00 | 0.98 | 0.9902 | 99.01 |
| AlexNet | 0.99 | 0.99 | 0.99 | 0.99 | 0.9902 | 99.01 |
| VGG19 | 0.98 | 1.00 | 1.00 | 0.98 | 0.9903 | 99.02 |
| ResNet50 | 0.99 | 1.00 | 1.00 | 0.99 | 0.9931 | 99.31 |
| **Proposed SSL work-frame** | **0.98** | **0.98** | **0.98** | **0.98** | **0.982** | **98.24** |

**Mammography** The mammography model (proposed) achieved test accuracy **98.24%**, F1-score **0.982**, and AUC **0.9987** across 1,020 balanced test images, with precision and recall of 0.98 for both classes. Near-identical performance across classes confirms oversampling prevented class bias. This is competitive with published supervised results on INbreast, achieved via pretraining requiring no labeled data.

**Performance Comparison with Baseline Models** The most clinically significant thermography observation is the asymmetry between malignant recall (0.86) and benign recall (0.57). In cancer screening, missing a malignant case carries greater consequences than a false alarm, so this sensitivity bias aligns with clinical priorities. The lower benign recall (0.57) could be improved via larger labeled datasets or threshold tuning. The F1-score of 0.7526 and AUC of 0.7816 confirm that the SSL framework extracted genuinely useful thermal features—substantially above the 50% random baseline in early training.

Tables 1 and 2 compare the proposed SSL workflow against four supervised baseline architectures.

On thermography, our SSL workflow achieves the highest accuracy (71.60%) and F1-score (0.7526), with malignant recall of 0.86 substantially exceeding all baselines (LeNet5: 0.70, AlexNet: 0.52, VGG19: 0.62, ResNet50: 0.67). On mammography, all models perform at a comparably high level (98.24%–99.31% accuracy); the SSL workflow's marginally lower accuracy relative to ResNet50 (99.31%) reflects the self-supervised pretraining objective's prioritization of representation generalizability, while supervised baselines may be overfit.

### 5.2 Ablation Study (Thermography Only)

Table 3 isolates the contributions of preprocessing and SSL pretraining within the thermography pipeline.

Table 3: Ablation study results for the thermography pipeline of Proposed SSL Model/Work-frame.

| Configuration | Precision | | Recall | | F1-score (pos.) | Acc. (%) |
|---|---|---|---|---|---|---|
| | Malignant | Benign | Malignant | Benign | | |
| Without preprocessing | 0.64 | 0.74 | 0.80 | 0.56 | 0.7142 | 67.90 |
| Without SSL (ResNet-50 only) | 0.70 | 0.68 | 0.67 | 0.72 | 0.6835 | 69.13 |
| **Full Proposed Work-frame (Preprocessing + SSL + head)** | **0.67** | **0.81** | **0.86** | **0.57** | **0.7526** | **71.60** |

Removing preprocessing reduces accuracy to 67.90% and malignant recall to 0.80, underscoring median filtering and CLAHE's diagnostic importance. Removing SSL entirely reduces accuracy to 69.13% and malignant recall to 0.67—matching the ResNet50 baseline in Table 1 and confirming internal consistency. The drop in malignant recall from 0.86 to 0.67 on SSL removal is the single largest degradation across all ablations, establishing SSL pretraining as the most consequential contributor given the limited 1,206-image training set.

### 5.3 Computational Efficiency

For the thermography SSL work-frame, total inference over 162 test images was **2.8248 s** (≈17.4 ms/image); for mammography over 1,020 test images: **5.0235 s** (≈4.9 ms/image). The mammography model processes images roughly 3.5× faster, attributable to the absence of Gabor and symmetry components. Both work-frames nonetheless demonstrate inference speeds well within the sub-second per-image range, supporting their suitability for real-time or near-real-time clinical screening workflows. This also compares favorably with Roslidar et al.'s [3] bi-pipeline SSL framework, which required ≈4 seconds for a single combined segmentation and classification inference pass, whereas the proposed thermography work-frame processes each image in under 18 ms.

### 5.4 Comparative Analysis

The 26.6 percentage-point gap between mammography (98.24%) and thermography (71.60%) is due to three structural differences between the modalities: *dataset scale* (8,160 vs. 1,206 training images, in our case); *signal clarity* (mammographic morphological detail vs. diffuse thermal vascular patterns susceptible to room temperature and patient positioning); and *label certainty* (biopsy-confirmed mammographic ground truth vs. inherently noisier thermographic labels).

Table 4 presents our thermography results against prior works. Despite the use of such small primary dataset for our proposed framework, the model showed potential as it outperformed former State-of-the-art models on f1-score (0.7526), and malignant recall (0.86) despite maintaining patient-level dataset split on the same dataset using the Self-supervised workflow. Though this framework currently lacks external validation on big datasets like DMR-IR like done by [6], cross validation via mammography dataset shows enough potential as well as generalizabilty in shape of dual-modality for now.

Table 4: Comparison of Related Thermographic Detection Studies with Our Study

| Study | Dataset | Model | Acc. | Pat. level Split | Limitation |
|---|---|---|---|---|---|
| [6] | Primary[15] (same as ours) | ***Thermo-CAD (CNN+NNMF)*** | *79.3%* | No | 100% accuracy on the DMR-IR dataset indicates overfitting; 79.3% accuracy on imbalanced dataset is misleading, their poor recall (0.5429) and f1-score on malignant class against ours shows poor generalization. |
| [8] | Primary (1000 images from 200 patients) | ***3D-BreastNet*** | *Dice-index, Hausdorff: [0.97, 3.86].* | **Yes** | Not a classification-based study (No classification method integrated). |
| Ours | Primary[15] | ***VICReg [ResNet-50 + Gated Fusion]*** | *71.60%* | **Yes** | Small thermographic dataset (357 images, 119 patients); no external thermographic validation, no cross-validation. |

## 6 Conclusion

This study presented a self-supervised framework with dual-modality for breast cancer detection, prioritizing thermography as a radiation-free, cost-accessible primary modality while using mammography as a secondary reference for validation while ensuring generalizability. The framework achieved 71.60% accuracy and AUC of 0.7816 on thermography (malignant recall: 0.86, F1: 0.7526) and 98.24% accuracy with AUC of 0.9987 on mammography, demonstrating that a single SSL architecture can serve both data-scarce thermographic and data-rich mammographic settings. Inference timings of ≈17.4 ms and ≈4.9 ms per image confirm practical deployability.

The main limitation is the small thermography dataset: 357 original images from 119 patients, with the test set of 162 images carrying meaningful statistical uncertainty. Future works must validate on larger thermography datasets; the strong mammography results provide supportive evidence of framework robustness, but thermography-specific external validation remains essential. Performance on lower-quality community-screening mammography images should also be explored, along with domain adaptation and multi-site training strategies. A more comprehensive component-level ablation independently isolating Gated Fusion, dual-pooling, and symmetry loss—and use of real unlabeled data for pretraining—would further strengthen generalizability.

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